

2011 年真题参考答案

一、选择题

(1) C. (2) B. (3) C. (4) C. (5) A. (6) B. (7) D. (8) D.

二、填空题

(9) $\sqrt{2}$. (10) $e^{-x} \sin x$. (11) $\ln(\sqrt{2} + 1)$. (12) $\frac{1}{\lambda}$. (13) $\frac{7}{12}$. (14) 2.

三、解答题

(15) $1 < \alpha < 3$.

(16) 极大值 $y(-1) = 1$, 极小值 $y\left(\frac{5}{3}\right) = -\frac{1}{3}$, 凹区间 $\left(\frac{1}{3}, +\infty\right)$, 凸区间 $\left(-\infty, \frac{1}{3}\right)$, 拐点 $\left(\frac{1}{3}, \frac{1}{3}\right)$.

(17) $f'_1(1, 1) + f''_{11}(1, 1) + f''_{12}(1, 1)$.

(18) $y = \arcsin \frac{e^x}{\sqrt{2}} - \frac{\pi}{4}$.

(19) 证明略.

(20) (I) $\frac{9\pi}{4}(\text{m}^3)$.

(II) $\frac{27 \times 10^3}{8} \pi g(\text{J})$.

(21) a .

(22) (I) $a = 5$.

(II) $\beta_1 = 2\alpha_1 + 4\alpha_2 - \alpha_3$, $\beta_2 = \alpha_1 + 2\alpha_2$, $\beta_3 = 5\alpha_1 + 10\alpha_2 - 2\alpha_3$.

(23) (I) 特征值 $-1, 1, 0$, 分别对应特征向量 $k_1(1, 0, -1)^T$, $k_2(1, 0, 1)^T$, $k_3(0, 1, 0)^T$, 其中 k_1, k_2, k_3 为任意非零常数.

(II) $\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$.